Is Poverty a Contributor to Low Literacy Rates?

A. Introduction and Summary of Research Question

Every two to five years, the National Center for Education Statistics, an organization within the U.S. Department of Education, conducts a thorough examination of student performance in various subjects known as the National Assessment of Educational Progress (NAEP) (National Center for Education Statistics, n.d.-b). The results are then published as "The Nation's Report Card" (National Center for Education Statistics, n.d.-b). In 2024, the organization reported the lowest reading scores among 12^{th} graders ever reported in an NAEP assessment (National Assessment Governing Board, 2024). This score continues a decline in senior reading scores that began in 1998 (National Center for Education Statistics, n.d.-a).

If educators and lawmakers aim to slow and eventually reverse this decline in literacy rates, they require concrete answers about the major contributing factors and data-backed solutions to promote literacy improvement. This report seeks to answer one primary question about the root causes of illiteracy in the United States: Is poverty a significant predictor? If a statistically significant link can be identified between poverty rate and literacy rate, it would provide valuable evidence that literacy interventions should be directed at low-income communities and that policy that reduces poverty would also improve literacy rates. It will be the goal of this report to establish this link using a linear regression model that will provide coefficients and significance for a variety of possible predictive factors, including poverty rate.

Formally, the identified research question will be as follows: Is the poverty rate of a county, defined as the proportion of residents who live below the poverty line, a statistically significant predictor of that county's average literacy score? The null hypothesis is "Poverty rate is not a statistically significant predictor of literacy score." The alternative hypothesis is "Poverty rate is a statistically significant predictor of literacy score." Data will be collected that estimates the average literacy score for a given county in the U.S., along with general demographic information about that county, including the proportion of residents below the poverty line. A linear regression model will be created that will predict literacy score as a function of many features in this data set, including the poverty rate. The linear regression process will provide coefficients, significance scores, and, most importantly, p-values for each predictor. The p-value found for the poverty rate predictor will be used to reject the null hypothesis or fail to reject it.

B. Data Gathering Process

The dataset that will be used in this analysis is the Skills Map dataset collected by the Programme for the International Assessment of Adult Competencies (PIAAC). The Skills Map was published on the website for the Department of Education, which included an option to download the data as a Microsoft Excel spreadsheet (PIAAC, 2023). The Skills Map provides estimates for the average literacy and numeracy scores for each county in the United States. It also provides various demographic information about each county, such as the population, the proportion of residents at different education levels, the proportion of residents in different types of occupations, the poverty rate, the proportion of residents receiving SNAP benefits, and more.

The downloaded dataset contained multiple ways of examining the data, many of which were not relevant for this analysis. For example, there were worksheets in the spreadsheet that examined the data on a per-state basis and on a national level. These being ill-suited to training a machine learning model, they were removed. The dataset also included ways to view the data for groups of residents according to their age and education level, but these were also ignored. As such, the data gathering processed was comprised of downloading the dataset, selecting only the worksheet with county-level data, and removing all records that represented subgroups within each county rather than each county as a whole.

One advantage of the data gathering methodology used for this analysis is the consistency that comes from using an already assembled data set. For example, the data was collected between 2012 and 2017, meaning that there are minimal time-based discrepancies in the dataset. The minimal discrepancies make the data gathering process convenient. If this analysis instead used data gathered from multiple sources, then the data would need to be verified with cross-validation, and issues would need to be rectified. Columns of data may also need to be converted to differing measures or data types to ensure all data is in the same format. Using the assembled dataset alleviates all of these issues.

The primary disadvantage of this methodology is the lack of supporting information. This results in an inability to consider other factors not included in the dataset, meaning that important information about the causes of low literacy might be missed. Also, since the data set has a fixed, somewhat small, set of features, there is no way to perform meaningful feature selection. Still, this disadvantage should not be terribly impactful on the analysis, as the data set is quite broad in terms of the variables it represents and is very well-suited to answering the research question.

The challenges encountered during this process were minimal. The most significant were determining how the data was formatted and extracting the relevant data. The former proved challenging since columns used obscure abbreviations and labels such as "Lit_P1" for the proportion of residents with an estimated literacy at or below Level 1. This issue was addressed by consulting the data dictionary included in a worksheet within the downloaded Excel spreadsheet. The latter challenge was that the data was stored in several different formats that needed to be parsed. For example, statistics were recorded for an entire county, but also for different subgroups in that county based on age and education level. To sort through this information, the Pandas library was used to examine the data and ultimately select only the records that recorded information for an entire county.

C. Data Extraction Process

The title of the data set downloaded from the Skills Map website was “SAE_website_dataset” followed by a string of assorted numbers and letters. The file was renamed by removing these latter excess characters. Then, within the Excel sheet were multiple worksheets; the one relevant to this analysis was titled “County.”

	A	B	C	D	E	F	G	H
1	FIPS_code	State	County	grpName	Lit_P1	Lit_P1_CI_L	Lit_P1_CI_U	Lit_P1_CV
2	1001	Alabama	Autauga Cour	all	0.205	0.169	0.241	0.086
3	1003	Alabama	Baldwin Coun	all	0.161	0.125	0.197	0.11
4	1005	Alabama	Barbour Coun	all	0.394	0.343	0.445	0.067
5	1007	Alabama	Bibb County	all	0.269	0.228	0.313	0.081
6	1009	Alabama	Blount County	all	0.248	0.204	0.29	0.087
7	1011	Alabama	Bullock Count	all	0.447	0.383	0.514	0.075
8	1013	Alabama	Butler County	all	0.32	0.274	0.368	0.074
9	1015	Alabama	Calhoun Cour	all	0.25	0.215	0.286	0.072
10	1017	Alabama	Chambers Co	all	0.299	0.254	0.347	0.079
11	1019	Alabama	Cherokee Cou	all	0.253	0.209	0.294	0.085
12	1021	Alabama	Chilton Count	all	0.262	0.219	0.305	0.085
13	1023	Alabama	Choctaw Cou	all	0.318	0.273	0.366	0.075
14	1025	Alabama	Clarke County	all	0.322	0.269	0.378	0.086
15	1027	Alabama	Clay County	all	0.318	0.266	0.369	0.083
16	1029	Alabama	Cleburne Cou	all	0.279	0.228	0.329	0.092
17	1031	Alabama	Coffee Count	all	0.224	0.189	0.259	0.077
18	1033	Alabama	Colbert Count	all	0.233	0.196	0.268	0.079
19	1035	Alabama	Conecuh Cou	all	0.355	0.304	0.407	0.074
20	1037	Alabama	Coosa County	all	0.32	0.263	0.375	0.089
21	1039	Alabama	Covington Co	all	0.243	0.204	0.281	0.081
22	1041	Alabama	Crenshaw Co	all	0.284	0.241	0.327	0.078
23	1043	Alabama	Cullman Coun	all	0.226	0.187	0.266	0.09
24	1045	Alabama	Dale County	all	0.229	0.19	0.27	0.088
25	1047	Alabama	Dallas County	all	0.37	0.313	0.431	0.081
26	1049	Alabama	DeKalb Count	all	0.319	0.272	0.367	0.076
27	1051	Alabama	Elmore Count	all	0.215	0.177	0.253	0.089
28	1053	Alabama	Escambia Cou	all	0.309	0.267	0.351	0.069

County | State | Nation | Variable List | +

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In Python, the Pandas library was used to load this worksheet as a dataframe, specifying the County worksheet as the one to be read. With the data loaded, the names and data types of each column were previewed using the info() method to identify possible candidate columns for use in the analysis.

```
data = pd.read_excel('SAE_website_dataset.xlsx', sheet_name='County')
```

```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 34562 entries, 0 to 34561
Data columns (total 92 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   FIPS_code              34562 non-null  int64
 1   State                  34562 non-null  str
 2   County                 34562 non-null  str
 3   grpName                34562 non-null  str
 4   Lit_P1                 34562 non-null  float64
 5   Lit_P1_CI_L            34562 non-null  float64
 6   Lit_P1_CI_U            34562 non-null  float64
 7   Lit_P1_CV              34562 non-null  float64
 8   Lit_P1_indicator       47 non-null     float64
 9   Lit_P1_CI_L_indicator  65 non-null     float64
 10  Lit_P1_CI_U_indicator  215 non-null    float64
 11  Lit_P1_CV_indicator    3990 non-null   float64
 12  Lit_P2                 34562 non-null  float64
 13  Lit_P2_CI_L            34562 non-null  float64
 14  Lit_P2_CI_U            34562 non-null  float64
 15  Lit_P2_CV              34562 non-null  float64
 16  Lit_P2_indicator       179 non-null    float64
 17  Lit_P2_CI_L_indicator  794 non-null    float64
 18  Lit_P2_CI_U_indicator  35 non-null     float64
 19  Lit_P2_CV_indicator    2136 non-null   float64
 20  Lit_P3                 34562 non-null  float64
 21  Lit_P3_CI_L            34562 non-null  float64
 22  Lit_P3_CI_U            34562 non-null  float64
 23  Lit_P3_CV              34562 non-null  float64
```

The candidate columns were the average literacy score and poverty rate, as well as several other variables that could plausibly be predictors of literacy score. Once identified, these candidate columns were extracted, and the rest discarded.

```
candidate_cols = ['FIPS_code', 'State', 'County', 'grpName', 'Lit_A', 'POP', 'POP_DOMAIN', 'Male', 'Female', 'White', 'Black',
                  'Hispanic', 'Asian', 'AIAN', 'NHPI', 'Other_race', 'Less_HS', 'HS', 'More_HS', 'FB', 'Poverty_100',
                  'Poverty_150', 'SNAP', 'Employed', 'Unemployed', 'Not_in_labor', 'OCC_Manage', 'OCC_Service', 'OCC_Sales',
                  'OCC_Military', 'OCC_Prod', 'No_Insurance']
```

```
data = data[candidate_cols]
```

Next began the data exploration process. Comparing the size of the dataframe to the number of unique FIPS codes revealed a discrepancy. There were significantly more records than there were locations.

```
len(data)
```

```
34562
```

```
len(data['FIPS_code'].unique())
```

```
3142
```

The reason for this is the "group name" column. As described in part B of this report, each record in this set could represent an entire county or specific subgroups within the county, as specified in the group name column. There were subgroups based on different age ranges and on different levels of education.

```
data['grpName'].unique()
```

```
<StringArray>
[                 'all',                 '16_24',                 '25_34',
                '35_44',                 '45_54',                 '55_64',
                '65_74', 'Less_than_high_school',           'High_school',
         'Some_college',   'Bachelors_or_higher']
Length: 11, dtype: str
```

Since there were no plans to incorporate the divide into the analysis, only the rows representing entire counties for the analysis were selected. The now redundant group name column was then dropped.

```
data = data.loc[data['grpName'] == 'all']
```

```
data['grpName'].unique()
```

```
<StringArray>
['all']
Length: 1, dtype: str
```

```
data = data.drop(columns='grpName')
```

When choosing which variables to include in the analysis, one issue that needed to be considered was that of multicollinearity, the correlation of multiple predictor variables that weakens linear regression models. One way to solve the problem of colinearity is to use a technique like principal component analysis; however, that technique obscures the direct effect of each predictor on the outcome variable, which would prevent the model from providing the information necessary to answer the research question.

Instead, this analysis used the technique of manual correlation analysis to identify columns that might contribute to colinearity in the model, and redundant columns were then removed. The technique used to identify correlated columns was Spearman's rank correlation coefficient. The simplicity of this technique meant that the correlation between two variables could be ascertained very quickly. The package Scipy was used to perform the test and report the correlations and corresponding p-values. One advantage of this technique is that it is able to identify nonlinear correlations, unlike other techniques such as Pearson's correlation coefficient. One disadvantage is its inability to examine multiple variables at once; rather, pairs of variables need to be considered one at a time.

There were two pairs of variables with nearly perfect correlation. The first pair was the county population and the "domain population," which is the number of residents between 16 and 74. The second was made up of the 100% poverty rate and the 150% poverty rate variables. These recorded the number of residents below 100% and 150% of the poverty line, respectively. Since these pairs were nearly perfectly correlated, one variable was dropped from each pair. The variables kept for this analysis were the total population and 100% poverty rates.

Population and domain population

```
scipy.stats.spearmanr(data['POP'], data['POP_DOMAIN'], nan_policy='omit')
```

```
SignificanceResult(statistic=np.float64(0.9996007092588998), pvalue=np.float64(0.0))
```

Near perfect correlation. Drop domain population.

```
data = data.drop(columns='POP_DOMAIN')
```

Two poverty rates

The dataset contains two poverty rates. Poverty 100% includes all people living below the poverty line. Poverty 150% includes all people living below 150% of the poverty line, meaning that it includes some people that live slightly above the poverty line.

```
scipy.stats.spearmanr(data['Poverty_100'], data['Poverty_150'])
```

```
SignificanceResult(statistic=np.float64(0.9489774661329514), pvalue=np.float64(0.0))
```

Unsurprisingly, these are highly correlated. For simplicity, we will keep the 100% poverty rate figure and drop the 150% one.

```
data = data.drop(columns='Poverty_150')
```

The next group of columns that were examined was those that could be mere proxies for poverty. The variables considered for this category were the education levels, the SNAP benefit rate, and the insurance enrollment rate. Because higher education both costs money and gates high-level jobs, it was entirely plausible that it would be highly correlated with the poverty rate. In the same manner, since insurance costs money, it is likely that the insurance rate would be negatively correlated with poverty. The reception of SNAP benefits is tied directly to someone's income, creating a clear link. Indeed, when all three of these variables were examined, a statistically significant correlation with poverty was found. As such, all were dropped, leaving the poverty rate variable to stand on its own.

Proxies for poverty

Education level, SNAP benefit rate, and insurance rate could all be proxies for low education areas since low income directly affects all three.

```
]: scipy.stats.spearmanr(data['Less_HS'], data['Poverty_100'])
]: SignificanceResult(statistic=np.float64(0.6737306939210644), pvalue=np.float64(0.0))
]: scipy.stats.spearmanr(data['HS'], data['Poverty_100'])
]: SignificanceResult(statistic=np.float64(0.24398111517835058), pvalue=np.float64(8.276081092250019e-44))
]: scipy.stats.spearmanr(data['More_HS'], data['Poverty_100'])
]: SignificanceResult(statistic=np.float64(-0.5564202720369873), pvalue=np.float64(6.2013088459937e-255))
```

Unsurprisingly, there is a correlation between education level and poverty rate.

```
]: data = data.drop(columns=['Less_HS', 'HS', 'More_HS'])
]: scipy.stats.spearmanr(data['SNAP'], data['Poverty_100'])
]: SignificanceResult(statistic=np.float64(0.8124303583794217), pvalue=np.float64(0.0))
]: scipy.stats.spearmanr(data['No_Insurance'], data['Poverty_100'])
]: SignificanceResult(statistic=np.float64(0.498503221728745), pvalue=np.float64(4.5593845530801746e-197))
]: data = data.drop(columns=['SNAP', 'No_Insurance'])
```

The final groups examined for colinearity were those that represented multiple classes within a category. For example, the “Employed,” “Not Employed,” and “Not in Labor” columns represent proportions of residents in each possible employment class, and all residents are accounted for across all three groups. This means that if two of these values are known, the third can be directly calculated. This introduces multicollinearity that must be rectified by dropping one of the possible classes. Accordingly, the “Male” class was dropped from among the gender categories, the “White” class was dropped from among the racial categories, and the “Not in labor” class was dropped from among the employment categories.

```
data = data.drop(columns=['Male', 'White', 'Not_in_labor'])
```

The strategy of dropping one class from the other helps to limit the colinearity, but the technique has advantages and disadvantages. The primary advantage of the technique is that it is very simple, making a significant impact on the colinearity without requiring a lot of code or additional analysis. The main disadvantage of the strategy is that its efficacy decreases as the number of classes in each category increases. That is, if you have three classes, you only need to drop one since the ability to calculate the proportion of the second class from only the first is impossible. If you have thirty classes, however, each individual class’s contribution becomes small, meaning that you can create a decent estimation of remaining classes even if multiple have been dropped. This disadvantage was considered for this analysis, but the largest category in the set only contained seven classes, and the other categories were significantly smaller. As such, this technique was deemed sufficient.

The last columns dropped were the identifier columns, which did not include meaningful information for the analysis. These were the FIPS code, state name, and county name columns.

Lastly, we should drop the FIPS_code, state, and county variables since these are simply identifiers and do not provide meaningful information for performing linear regression.

```
data = data.drop(columns=['FIPS_code', 'State', 'County'])
```

This left the following variables to be those used in the analysis:

- Average literacy score (the outcome variable)
- Population
- Proportion of residents who are female
- Proportion of residents who are black
- Proportion of residents who are Hispanic
- Proportion of residents who are Asian
- Proportion of residents who are American Indian/Alaskan Native
- Proportion of residents who are Native Hawaiian/Pacific Islander
- Proportion of residents who are of another non-white race
- Proportion of residents who are foreign-born
- Proportion of residents who are below the poverty line
- Proportion of residents who are employed
- Proportion of residents who are in the labor force but are currently unemployed
- Proportion of residents who are in managerial occupations
- Proportion of residents who are in service occupations
- Proportion of residents who are in sales occupations
- Proportion of residents who are in military occupations
- Proportion of residents who are in production occupations

```
data.columns
```

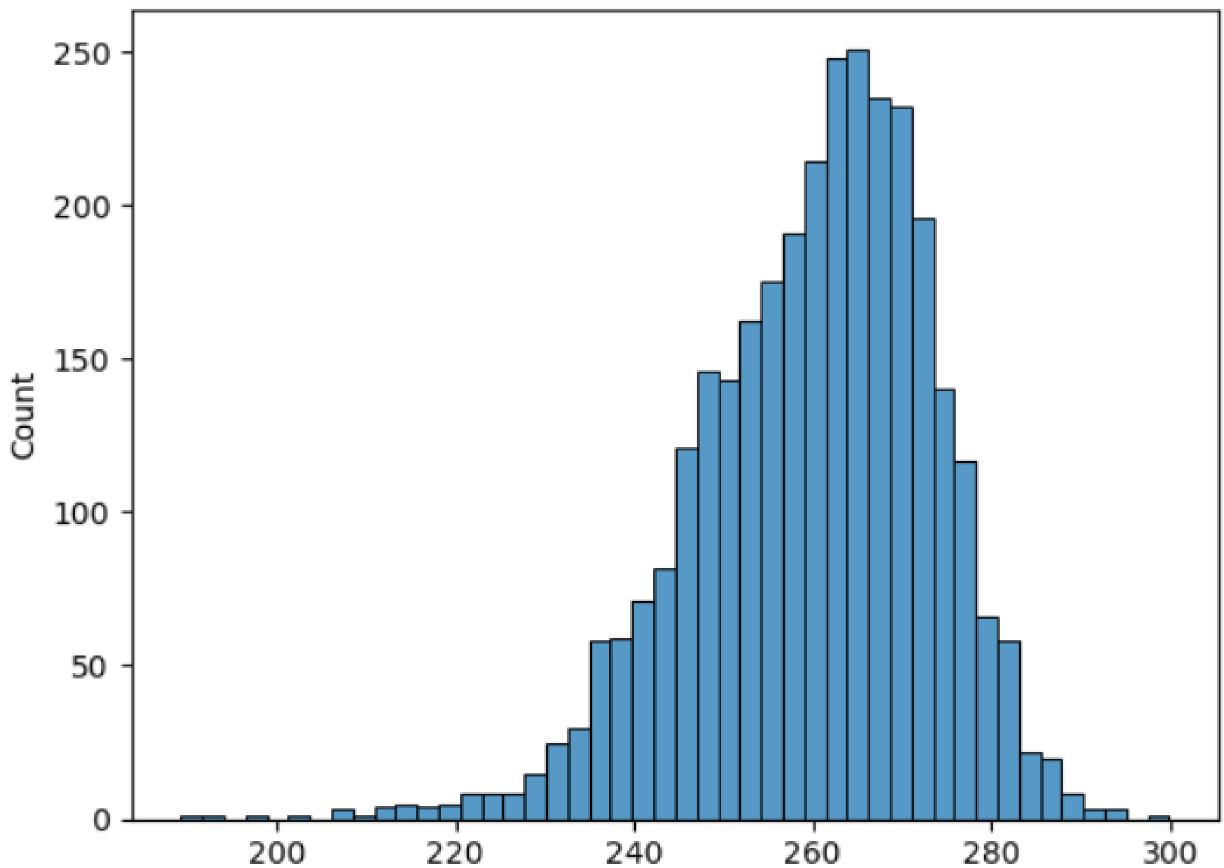
```
Index(['Lit_A', 'POP', 'Female', 'Black', 'Hispanic', 'Asian', 'AIAN', 'NHPI',
       'Other_race', 'FB', 'Poverty_100', 'Employed', 'Unemployed',
       'OCC_Manage', 'OCC_Service', 'OCC_Sales', 'OCC_Military', 'OCC_Prod'],
```

The next step in the data preparation process was to explore the data while searching for missing values and outliers. To accomplish both steps at the same time, a function was created in Python that could analyze each column for its measures of central tendency, variance, skew, minimum, and maximum. It would also count missing values and identify outliers according to the data's interquartile range.

```python
def summarize(series: pd.Series) -> None:

    print('=' * max(10, len(series.name)))
    print(series.name)
    print(f'Data type: {series.dtype}')
    if series.isna().sum() != 0:
        print(f'Missing values: {series.isna().sum()}')

    print(f'Mean: {series.mean()}')
    print(f'Median: {series.median()}')
    print(f'Standard deviation: {series.std()}')
    print(f'Variance: {series.var()}')
    print(f'Skew: {series.skew()}')
    print(f'Max: {series.max()}')
    print(f'Min: {series.min()}')
    q1 = np.percentile(series, 25)
    q3 = np.percentile(series, 75)
    iqr = q3 - q1
    low = q1 - 1.5 * iqr
    high = q3 + 1.5 * iqr
    outlier_count = (series < low).sum() + (series > high).sum()
    if outlier_count != 0:
        print(f'Outliers: {outlier_count}')

    print('=' * max(7, len(series.name)))

    plt.figure()
    sns.histplot(series)
    plt.show()
```

This summarize() function was used on every single variable in the dataset. For brevity, only a sample will be included here. The following is the summary of the analysis of the average literacy variable.

```
==========
Lit_A
Data type: float64
Mean: 259.9894334818587
Median: 261.7
Standard deviation: 13.328660465683965
Variance: 177.6531898094867
Skew: -0.6516374989833967
Max: 299.9
Min: 189.3
Outliers: 43
=======
```



The summaries reported no missing values anywhere in the data set, but they did report some outliers. Importantly, outliers are not necessarily invalid by default. They simply need to be examined further to determine if they are valid values.

The technique used to identify outliers was to use the maximum and minimum values in each category and compare them to reasonable ranges. For example, the variables representing proportions must be within 0 and 1, so if the minimum and maximum are within those bounds, then all of the outliers are within theoretically valid ranges. One advantage of this strategy is its ability to leverage domain knowledge to quickly classify whether outliers are valid or invalid. In the case of proportions, there are hard limits on the values that can be taken: they must be between 0 and 1. In other cases, however, an analyst can still use domain knowledge to identify reasonable ranges and use the technique accordingly. For example, when analyzing the outliers for the population variable, domain knowledge can be used to quickly determine that the maximum and minimum observed populations are of a reasonable size. According to the Census Bureau's 2025 estimates, the least and most populous counties in the U.S. have 52 and nearly ten million residents, respectively (United States Census Bureau, 2025). These are comparable to the minimum and maximum found in this dataset from 2023 and are therefore likely valid.

```
==========
POP
Data type: float64
Mean: 102165.62921705919
Median: 25691.5
Standard deviation: 328292.19514528924
Variance: 107775765393.31265
Skew: 13.925722377382574
Max: 10105722.0
Min: 74.0
Outliers: 425
=======
```

Using this strategy, it was determined that there were no obvious outliers in the dataset to be removed.

The next step in the data preparation process was the standardization of the predictor variables. While this was not strictly necessary for the analysis, it does provide the advantage of making the linear coefficients of the model more easily comparable to one another. Without standardization, the coefficients of variables with larger values on average will naturally be smaller even if their contribution to the predicted outcome variable is larger than that of another variable. The primary advantage of standardization is that it scales all variables down to the same range, making the size of the coefficients a more reasonable proxy for the significance each variable plays in the prediction. The primary disadvantage is that it obscures the actual values for each column, which means that the coefficients cannot be used directly on raw values to estimate the true contribution of each variable to the outcome for a particular example. Since this is not necessarily a concern for this particular analysis, the approach was deemed appropriate.

```
def standardize(series: pd.Series) -> pd.Series:
    mean = series.mean()
    std = series.std()
    return (series - mean) / std
```

```
for column in data.columns:
    if column == 'Lit_A':
        continue
    data[column] = standardize(data[column])
```

The final step in the preparation process was to rename the variables to something more consistent, readable, and understandable. Capital letters were lowercased, abbreviations were expanded to include whole words, and some variables had their names re-ordered or rewritten to ensure clarity.

```
data = data.rename(columns=lambda name: name.lower())
```

```
data = data.rename(columns={
    'lit_a': 'avg_literacy',
    'pop': 'population',
    'aian': 'native_american',
    'nhpi': 'native_hawaiian_pacific_islander',
    'fb': 'foreign_born',
    'poverty_100': 'below_poverty_line',
    'occ_manage': 'managerial_occupation',
    'occ_service': 'service_occupation',
    'occ_sales': 'sales_occupation',
    'occ_military': 'military_occupation',
    'occ_prod': 'production_occupation'
})
```

The completely prepared data was then saved to a file called “prepared_data.csv” to be used for further analysis.

```
data.to_csv('prepared_data.csv', index=False)
```

D. Analysis

The analysis technique selected for this analysis was Linear Regression. This technique is appropriate for addressing the research question, as it will allow for a variety of possible predictors to be considered when trying to determine if there is a significant link between poverty rate and literacy rate specifically. Additionally, many statistical packages, including StatsModels, the one used for this analysis, will provide p-values for each coefficient, which test if the coefficient is nonzero. This provides precisely the information necessary to establish statistical significance and answer the research question.

One advantage of the linear regression technique is its interpretability. The model provides coefficients, t-statistics, and p-values for every predictor, denoting its relationship with the outcome variable. This provides ample information that can be used to draw conclusions about the data. One disadvantage of using this technique is that it can only model linear relationships. As such, if there is a nonlinear relationship between the predictor and outcome variable, the model must do its best to estimate a linear one, which may make the coefficients and predictions less accurate. Nonetheless, the aforementioned advantages make the technique very well-suited for addressing this particular research question.

The process of forming the model began with dividing the data into training and testing sets. While this is not strictly required for an explanatory analysis, being able to test the model's predictions on a holdout set could still prove useful for determining the model's accuracy. The data split was performed using Scikit-Learn's train_test_split() function and used 80% of the data for the training set and 20% for the testing set.

```
prepared_data = pd.read_csv('prepared_data.csv')
X = prepared_data.drop(columns='avg_literacy')
y = prepared_data['avg_literacy']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0, shuffle=True)
```

```
X_train.head()
```

	population	female	black	hispanic	asian	native_american	native_hawaiian_pacific_islander	other_race	foreign_b
1767	-0.213367	-1.081414	-0.456748	-0.541278	-0.266094	-0.201630	0.008458	-0.529748	-0.175
113	-0.186031	0.736499	-0.608358	-0.512104	-0.337915	-0.188592	-0.127127	-0.418250	-0.439
1323	-0.010255	0.158072	-0.518770	-0.366237	0.452116	-0.227708	-0.127127	-0.329051	0.017
1504	-0.058294	0.653866	-0.580792	-0.461051	-0.302005	-0.188592	0.008458	-0.329051	-0.562
2955	0.269161	-0.007193	-0.518770	0.851754	0.416205	-0.149475	0.008458	2.302309	0.982

To create the actual model, StatsModel's OLS object was used. When the OLS object is constructed, it expects the data to be passed in with a column of constants. The StatsModels add_constant() function creates that column in data sets that currently lack one.

```
X_train_sm = sm.add_constant(X_train)
X_test_sm = sm.add_constant(X_test)
```

Next, the model was created by passing the training data into the OLS() constructor, and the fit() method was called on the resulting model to perform the regression.

```
model = sm.OLS(y_train, X_train_sm)
```

```
fit = model.fit()
```

Finally, the summary method was called on the fitted model, which provides the statistics necessary to answer the research question.

OLS Regression Results

Dep. Variable:	avg_literacy	**R-squared:**	0.881
Model:	OLS	**Adj. R-squared:**	0.881
Method:	Least Squares	**F-statistic:**	1091.
Date:	Tue, 21 Apr 2026	**Prob (F-statistic):**	0.00
Time:	21:29:31	**Log-Likelihood:**	-7425.5
No. Observations:	2513	**AIC:**	1.489e+04
Df Residuals:	2495	**BIC:**	1.499e+04
Df Model:	17		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	259.9974	0.093	2793.144	0.000	259.815	260.180
population	-0.1221	0.106	-1.151	0.250	-0.330	0.086
female	-0.0745	0.121	-0.614	0.539	-0.312	0.163
black	-3.7722	0.118	-31.963	0.000	-4.004	-3.541
hispanic	-4.9669	0.163	-30.392	0.000	-5.287	-4.646
asian	0.6898	0.171	4.023	0.000	0.354	1.026
native_american	-0.4985	0.113	-4.415	0.000	-0.720	-0.277
native_hawaiian_pacific_islander	-0.2092	0.193	-1.084	0.278	-0.588	0.169
other_race	0.5793	0.122	4.759	0.000	0.341	0.818
foreign_born	-0.6154	0.193	-3.190	0.001	-0.994	-0.237
below_poverty_line	-3.2120	0.170	-18.907	0.000	-3.545	-2.879
employed	3.2053	0.172	18.668	0.000	2.869	3.542
unemployed	0.6526	0.127	5.150	0.000	0.404	0.901
managerial_occupation	5.9329	0.196	30.281	0.000	5.549	6.317
service_occupation	0.9345	0.131	7.130	0.000	0.678	1.192
sales_occupation	1.7538	0.114	15.410	0.000	1.531	1.977
military_occupation	1.8562	0.098	18.854	0.000	1.663	2.049
production_occupation	0.4767	0.184	2.591	0.010	0.116	0.837

The R^2 of 0.881 indicates a fairly well-fit linear model. Later in the summary, the rounded p-value for the poverty rate coefficient is listed as 0.000. The exact value can be examined by referencing the fit model's p-values attribute.

```
fit.pvalues.loc['below_poverty_line']
```

```
np.float64(1.2948732858750045e-74)
```

Doing so lists the p-value as 1.29×10^{-74}. This is extremely close to zero, and much smaller than a standard significance level of 0.05. This conclusively indicates that the proportion of residents below the poverty line is a statistically significant predictor of a county's average literacy score.

To further confirm the model's validity, it was tested on holdout data. First, the testing set was passed into the fitted model's predict() function to get a list of predictions. Using

StatsModel's mse() function, the mean squared error and the R-squared value on the testing data were calculated.

```
y_pred = fit.predict(X_test_sm)
```

```
mse(y_test, y_pred)
```

```
np.float64(19.994971418059162)
```

```
def r_squared(y_test, y_pred) -> float:
    return 1 - mse(y_test, y_pred) / mse(y_test, y_test.mean())
```

```
r_squared(y_test, y_pred)
```

```
np.float64(0.8749024521945568)
```

The model achieved a mean squared error of about 20. This is not an exceptional predictive performance given that the range of possible scores runs from about 190 to 300, but while not exceptional, it is respectable. Luckily, predictive power is not an important goal of this analysis. The R^2 score achieved by the model on the testing data was about 0.875, which is very close to the training score of 0.881. This indicates that the model is not overfit to the training data and that the model's analysis of the training data should generalize to unseen data points.

E. Results and Reccomendations

The regression model identified a p-value of nearly zero for the coefficient between the proportion of residents below the poverty line and the average literacy score of a county. This provides ample evidence that there is a meaningful relationship between the two variables. The p-value being well below a 5% significance level means that the null hypothesis, "Poverty rate is not a statistically significant predictor of literacy score," can be safely rejected in favor of the alternative.

One limitation of the analysis is that linear regression cannot prove direct causation. Linear regression simply models the relationship between variables rather than establishing directional links. Also, while an effort was made to control for confounding factors, there will likely be some important factors missing. For these reasons, the analysis does not establish that interventions that alter the poverty rate will necessarily have a direct linear effect on the literacy of a county. Still, it does provide evidence of some general relationship and may be sufficient justification for experimenting with various policies that reduce the poverty rate with a close examination of the resulting effects.

Accordingly, the course of action recommended is that a comprehensive literacy improvement plan be made at the national level, which should include, among many other literacy improvement strategies, targeted interventions for low-income areas and interventions that attempt to address poverty directly. The findings of this report indicate that this could be an effective means of improving literacy. However, since the report cannot prove causality, the improvement plan should also include frequent monitoring of literacy rates year over year to determine the effectiveness of these interventions and determine a plan for modifying the interventions accordingly.

One recommendation for further study of the dataset would be to improve the model's explanatory power by including polynomial and interaction terms. The model created for this analysis achieved an R^2 of about 0.88, which is respectable, but it also leaves much room for improvement. Introducing polynomial or other transformations of the variables would allow for the model to identify nonlinear relationships more effectively, and adding interaction terms would allow the model to identify the complex relationships between multiple predictors and the outcome variable. Of course, introducing many additional terms expands the number of predictors greatly, so such an analysis should also use some kind of feature selection to optimize the set of variables being considered for the model. If done so, both the predictive and explanatory power of the model should be increased.

Another direction for further analysis of this dataset would be to perform a similar analysis to the one done in this report, but for the numeracy score instead of the literacy score. Numeracy, the mathematical skill of citizens, is also a priority for educational and governmental institutions. While one would expect the numeracy and literacy rates to be connected and to result in similar findings, confirming this similarity or identifying the ways in which the two differ might provide useful information to guide how a broader educational improvement plan may be developed.

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